# MySQL Setup Guide

How to connect the Sentiment Analysis application to MySQL

Switching from **H2 (default)** to **MySQL**

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## 1. Overview: H2 vs MySQL

By default, the application uses an **H2 file-based database** which requires zero setup. For production or shared environments, you can switch to **MySQL** by activating the `mysql` Spring profile. **No code changes are needed** — it is purely a configuration switch.

**H2 (Default)**

```
# No setup needed
# Data stored in ./data/sentimentdb
```

**MySQL**

```
# Requires MySQL 8.0+ running
# Data stored in MySQL server
```

```
# H2 Console at /h2-console
# Good for: dev, demos, single-user

java -jar sentiment-analysis.jar
```

```
# H2 Console disabled
# Good for: production, multi-user

java -jar sentiment-analysis.jar \
  --spring.profiles.active=mysql
```

## 2. Prerequisites

| REQUIREMENT | DETAILS |
| --- | --- |
| **MySQL Server** | Version 8.0 or later, running and accessible |
| **MySQL User** | A user with privileges to create databases and tables |
| **Network Access** | The application must be able to reach the MySQL host and port |

ⓘ The MySQL JDBC driver ( `mysql-connector-j` ) is already included in the application's dependencies. No additional libraries need to be installed.

## 3. Quick Start (3 Steps)

If MySQL is already running on `localhost:3306` with user `root` / password `root`, you only need to do this:

1 **Ensure MySQL is running**

```
mysql -u root -p -e "SELECT VERSION();"
```

2 **Set the Spring profile to `mysql`**

```
java -jar target/sentiment-analysis-0.0.1-SNAPSHOT.jar \
  --spring.profiles.active=mysql
```

Or via environment variable:

```
export SPRING_PROFILES_ACTIVE=mysql
java -jar target/sentiment-analysis-0.0.1-SNAPSHOT.jar
```

3 **Done!**

The application will automatically create the `sentimentdb` database and all tables on first startup.

✓ That's it for the default setup! If your MySQL uses a different host, port, username, or password, continue to the next section to configure environment variables.

## 4. Environment Variables

The MySQL profile uses environment variables with sensible defaults. Override only the ones that differ from your setup:

| VARIABLE | DEFAULT | DESCRIPTION |
|---|---|---|
| `DB_HOST` | `localhost` | MySQL server hostname or IP address |
| `DB_PORT` | `3306` | MySQL server port |
| `DB_NAME` | `sentimentdb` | Database name (auto-created if it doesn't exist) |
| `DB_USERNAME` | `root` | MySQL username |
| `DB_PASSWORD` | `root` | MySQL password |

### Example: Custom MySQL Connection

```
# Set environment variables
export DB_HOST=192.168.1.100
export DB_PORT=3306
export DB_NAME=sentiment_prod
```

```
export DB_USERNAME=sentiment_user
export DB_PASSWORD=MySecurePassword123

# Run with MySQL profile
java -jar target/sentiment-analysis-0.0.1-SNAPSHOT.jar \
  --spring.profiles.active=mysql
```

### Single-line Alternative

```
DB_HOST=192.168.1.100 DB_USERNAME=sentiment_user DB_PASSWORD=MySecurePassword123
  java -jar target/sentiment-analysis-0.0.1-SNAPSHOT.jar \
  --spring.profiles.active=mysql
```

## 5. Configuration File Details

The MySQL configuration lives in `src/main/resources/application-mysql.yml`. This file is activated only when the `mysql` profile is enabled. Here is the complete file:

```
# application-mysql.yml
spring:
  datasource:
    url: jdbc:mysql://${DB_HOST:localhost}:${DB_PORT:3306}/${DB_NAME:sentimentdb
    username: ${DB_USERNAME:root}
    password: ${DB_PASSWORD:root}
    driver-class-name: com.mysql.cj.jdbc.Driver
  jpa:
    properties:
      hibernate:
        dialect: org.hibernate.dialect.MySQLDialect
  h2:
    console:
      enabled: false
```

### What Each Setting Does

| SETTING | PURPOSE |
| --- | --- |
| `createDatabaseIfNotExist=true` | Automatically creates the database on first connection if it doesn't exist |

| SETTING | PURPOSE |
| --- | --- |
| `useSSL=false` | Disables SSL for local connections (enable for production, see section 10) |
| `allowPublicKeyRetrieval=true` | Allows RSA public key retrieval for `caching_sha2_password` auth (MySQL 8 default) |
| `serverTimezone=UTC` | Prevents timezone mismatch errors between JVM and MySQL |
| `MySQLDialect` | Tells Hibernate to generate MySQL-compatible SQL |
| `h2.console.enabled=false` | Disables the H2 web console (not needed with MySQL) |

⚠ **Do not edit `application.yml`** to switch databases. The profile-based approach (`application-mysql.yml`) keeps both configurations clean and lets you switch with a single flag.

## 6. MySQL Server Setup

### Install MySQL

| PLATFORM | COMMAND |
| --- | --- |
| **macOS** (Homebrew) | `brew install mysql && brew services start mysql` |
| **Ubuntu / Debian** | `sudo apt install mysql-server && sudo systemctl start mysql` |
| **Windows** | Download MySQL Installer from `dev.mysql.com/downloads/installer/` |
| **Docker** | See Docker command below |

### Docker (Recommended for Quick Setup)

```
# Start MySQL in Docker
docker run -d \
  --name sentiment-mysql \
  -e MYSQL_ROOT_PASSWORD=root \
  -e MYSQL_DATABASE=sentimentdb \
  -p 3306:3306 \
  mysql:8.0

# Then run the application
java -jar target/sentiment-analysis-0.0.1-SNAPSHOT.jar \
  --spring.profiles.active=mysql
```

### Create a Dedicated MySQL User (Recommended)

Instead of using `root`, create a dedicated user for the application:

```
-- Connect to MySQL as root
mysql -u root -p

-- Create the database
CREATE DATABASE IF NOT EXISTS sentimentdb
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_unicode_ci;

-- Create a dedicated user
CREATE USER 'sentiment_user'@'%'
  IDENTIFIED BY 'YourStrongPassword';

-- Grant privileges only on the sentiment database
GRANT ALL PRIVILEGES ON sentimentdb.* TO 'sentiment_user'@'%';
FLUSH PRIVILEGES;
```

ⓘ The `utf8mb4` character set is essential for storing Dravidian scripts (Tamil, Malayalam, Kannada, Telugu) correctly. MySQL 8 uses `utf8mb4` by default.

## 7. Run Commands

### Option A: Command-line Flag

```
# Using the JAR
java -jar target/sentiment-analysis-0.0.1-SNAPSHOT.jar --spring.profiles.active=

# Using Maven
./mvnw spring-boot:run -Dspring-boot.run.profiles=mysql
```

### Option B: Environment Variable

```
export SPRING_PROFILES_ACTIVE=mysql
java -jar target/sentiment-analysis-0.0.1-SNAPSHOT.jar
```

### Option C: With All Custom Variables

```
export SPRING_PROFILES_ACTIVE=mysql
export DB_HOST=db.example.com
export DB_PORT=3306
export DB_NAME=sentimentdb
export DB_USERNAME=sentiment_user
export DB_PASSWORD=YourStrongPassword
export ADMIN_PASSWORD=MyAdminPass123

java -jar target/sentiment-analysis-0.0.1-SNAPSHOT.jar
```

### Option D: IntelliJ IDEA

In your Run Configuration:

1. Open **Run > Edit Configurations**
2. Select your Spring Boot configuration
3. In **Active profiles**, enter: `mysql`
4. In **Environment variables**, add: `DB_USERNAME=sentiment_user;DB_PASSWORD=YourPassword`
5. Click **Apply** and **Run**

## 8. Verify the Connection

1 **Check the startup logs**

Look for this line in the application output:

```
HikariPool-1 - Starting...
HikariPool-1 - Added connection com.mysql.cj.jdbc.ConnectionImpl@...
HikariPool-1 - Start completed.
```

If you see `HikariPool` starting with a MySQL connection, it's working.

### 2 Verify tables were created

```
mysql -u root -p sentimentdb -e "SHOW TABLES;"

+------------------------+
| Tables_in_sentimentdb  |
+------------------------+
| app_user               |
| prediction_record      |
+------------------------+
```

### 3 Verify the admin account

```
mysql -u root -p sentimentdb -e "SELECT username, role FROM app_user;

+----------+-------+
| username | role  |
+----------+-------+
| admin    | ADMIN |
+----------+-------+
```

### 4 Test the application

Open `http://localhost:8080` in a browser, log in with `admin` / `Admin@12345`, and run a sentiment analysis.

## 9. Migrating Data from H2 to MySQL

If you have existing data in the H2 database and want to move it to MySQL:

1. **Export data from H2**

   Open the H2 console at `http://localhost:8080/h2-console` (with the default H2 profile) and run:

   ```
   SELECT * FROM APP_USER;
   SELECT * FROM PREDICTION_RECORD;
   ```

   Export the results as CSV or SQL INSERT statements.

2. **Start the app with MySQL profile**

   This creates empty tables in MySQL.

   ```
   java -jar target/sentiment-analysis-0.0.1-SNAPSHOT.jar \
     --spring.profiles.active=mysql
   ```

3. **Import data into MySQL**

   Run your exported INSERT statements against MySQL:

   ```
   mysql -u root -p sentimentdb < exported_data.sql
   ```

⚠ The admin account is recreated on every startup if it doesn't exist, so you don't need to migrate it. User accounts and prediction records are the data to migrate.

## 10. Production Recommendations

| AREA | RECOMMENDATION |
| --- | --- |
| **SSL** | Change `useSSL=false` to `useSSL=true&requireSSL=true` in `application-mysql.yml` for encrypted connections. |
| **Dedicated user** | Never use `root` in production. Create a user with only the necessary privileges (see section 6). |
| **Password security** | Always set `DB_PASSWORD` and `ADMIN_PASSWORD` via environment variables or a secrets manager. Never use the defaults. |
| **DDL strategy** | Consider changing `spring.jpa.hibernate.ddl-auto` from `update` to `validate` in production, and manage schema changes with a migration tool (Flyway or Liquibase). |
| **Connection pool** | Spring Boot uses HikariCP by default with 10 connections. Adjust `spring.datasource.hikari.maximum-pool-size` based on load. |
| **Character set** | Ensure the MySQL database uses `utf8mb4` for full Unicode support (required for Dravidian scripts). |
| **Backups** | Set up regular `mysqldump` backups: `mysqldump -u root -p sentimentdb > backup.sql` |

## 11. Troubleshooting

| ERROR | CAUSE & FIX |
| --- | --- |
| `Communications link failure` | MySQL is not running or not reachable. Check: `mysql -u root -p -e "SELECT 1;"`<br>Verify the host, port, and firewall rules. |
| `Access denied for user 'root'@'localhost'` | Wrong username or password. Verify `DB_USERNAME` and `DB_PASSWORD` environment variables match your MySQL credentials. |

| ERROR | CAUSE & FIX |
| --- | --- |
| `Unknown database 'sentimentdb'` | The `createDatabaseIfNotExist=true` flag should handle this. If it fails, create manually: `CREATE DATABASE sentimentdb;` |
| `Public Key Retrieval is not allowed` | The `allowPublicKeyRetrieval=true` flag is already set. If you see this, ensure you're using the `mysql` profile, not the default. |
| `The server time zone value 'XXX' is unrecognized` | The `serverTimezone=UTC` flag handles this. If it persists, set MySQL's timezone: `SET GLOBAL time_zone = '+00:00';` |
| `Table 'sentimentdb.app_user' doesn't exist` | Tables should be auto-created by Hibernate ( `ddl-auto: update` ). If not, check that the MySQL user has CREATE TABLE privileges. |
| Application starts but uses H2 | The `mysql` profile is not active. Verify with: check the logs for `H2` vs `MySQL` in datasource URL. Add `--spring.profiles.active=mysql` . |
| Dravidian characters show as `????` | Database not using `utf8mb4` . Fix: `ALTER DATABASE sentimentdb CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;` |

ⓘ To switch back to H2, simply run the application without the `mysql` profile: `java -jar sentiment-analysis.jar`

## Summary: What to Change

Here's a quick checklist of everything needed to connect to MySQL:

| # | WHAT | CHANGE | REQUIRED? |
| --- | --- | --- | --- |
| 1 | Spring profile | Set `--spring.profiles.active=mysql` or `SPRING_PROFILES_ACTIVE=mysql` | **Yes** |

| # | WHAT | CHANGE | REQUIRED? |
|---|---|---|---|
| 2 | MySQL server | Must be running and accessible | **Yes** |
| 3 | `DB_HOST` | Set if MySQL is not on `localhost` | If remote |
| 4 | `DB_PORT` | Set if MySQL is not on port `3306` | If non-default |
| 5 | `DB_NAME` | Set if you want a different database name | Optional |
| 6 | `DB_USERNAME` | Set to your MySQL username | If not `root` |
| 7 | `DB_PASSWORD` | Set to your MySQL password | If not `root` |

✓ **No code changes, no file edits, no additional dependencies needed.** The `mysql` profile and environment variables handle everything.